Darius Chitoroaga

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Education

University College London London, UK
Bachelor's Degree in Computer Science *September 2022 — June 2026*
- Computer Systems, DSA, Supervised Learning, (Co)-Algebraic Structures, Robotics

Ashlawn School Rugby, UK
A-levels: Mathematics, Physics, Computer Science, Further Mathematics, EPQ *September 2020 — June 2022*
- Grades: A*A*A*AB

Work Experience

Software Engineer Intern June 2025 — September 2025
Meta *London, UK*
- Extracted core functionality from an internal Rust CLI tool and exposed it via a scalable API endpoint, cutting update cycles from 90 days to 6 hours.
- Designed and implemented an MCP server enabling internal AI tools to automatically lease and control devices, streamlining internal workflows.
- Implemented user-requested features in CLI tool increasing internal usage by 24%.
- Optimised API calls in internal tool from 8 to 2, reducing overall time to execute by 21%.

Software Engineer, IXN Placement September 2023 — March 2024
Avanade + UCL *London, UK*
- Devised new methods of solving problems with LLMs, leveraging decomposition of tasks (chain-of-thought).
- Designed full-stack legal "copilot" web app using LangChain, Next.js, Django.
- Managed Azure deployments, containerisation of applications (Docker Compose).
- Acted as project manager to my team, setting achievable goals and deliverables.

Software Engineer Intern Jun 2023 — September 2023
Cambridge Sensoriis *Cambridge, UK*
- Created a 3D simulator in Python for R&D, such as testing various Kalman Filter algorithms I wrote to attain more accurate and consistent results in real world. (Accuracy improvement of over 50%).
- Wrote device drivers, for an IMU device to transmit up to 8 d.p. accuracy accelerometer and gyroscope data to a main board.
- Created 3D visualiser for IMU data showing orientation and acceleration of device in an easily digestible manner, resulting in identification of over 5 serious bugs by various team members.

Projects

Pure + Functional Operating System, NixOS Configuration April 2024 — Present
- Maintain a personal configuration of NixOS + home-manager to create a completely pure/functional workflow.
- Experiment with upstream packages/features leading to open-source PRs.
- Constantly iterating to perfect my workflow/dev environment.
- Allows me to always fall back to working state when experimenting and take my workflow to any machine.
- Includes custom front-end shell in Typescript using GTK widgets (app launcher, status bar, notifications, etc.).

Autonomous Network Anomaly Detection, PeerWatch September 2025 — April 2026
- Daemon fingerprints subnet devices via nmap + passive packet capture; scores identity drift across 20+ weighted signals
- Fleet correlator detects coordinated attacks across peers; triggers LLM investigation agent with rule-based fallback and autonomous iptables remediation

Skills

- **Programming Languages**: Python, C, Rust, Java, JS, Go, Haskell, Bash/Zsh/Fish
- **Technologies**: React, Git, UNIX, Docker, Linux, LangChain, Embedded Systems, IoT
- **Other**: Poker, Tennis, Basketball, Neovim, Nix